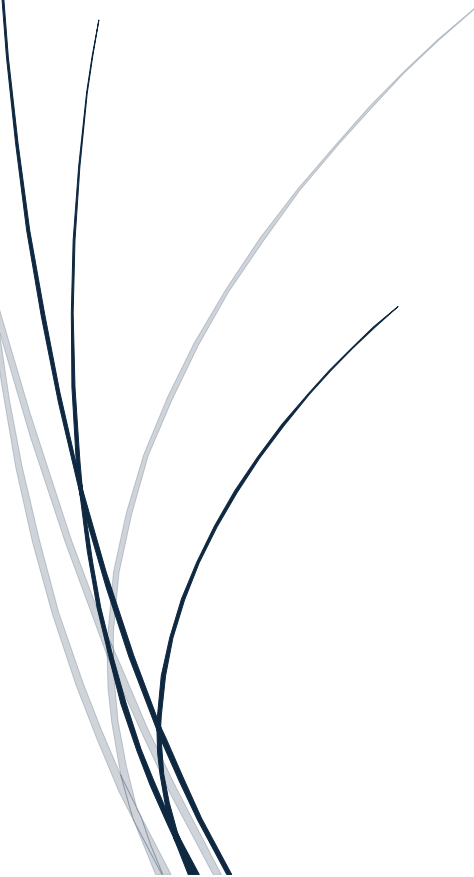


3/27/2025

Cricket Match Analysis Tool with Scheduling Algorithms

Operating System Assignment 2 (Q-3)



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Table of Contents

Method of Data Calculation	2
Screenshots	2
Bowlers	3
Batters	4
Predicted Run Scores and Wickets	4
Bowlers	4
Using RR.....	4
Using SJF	4
Batters	5
Using SRTF	5
Using SJF	5
Comparison of Each Scheduling Strategy	5
Bowlers	5
Using RR.....	5
Using SJF	6
Batters	6
Using SRTF	6
Using SJF	6
Recommendations	6
Bowlers	6
Batters.....	6

Method of Data Calculation

- The data was imported from a CSV file and calculated in the C++ code.
- Then it was again updated on the original CSV file.
- Each player's separate data was also updated on their separate CSV files.

Screenshots

```
Choose the scheduling type for batter:
1. SRTF
2. SJF
Choose: 1

Choose the scheduling type for bowler:
1. RR
2. SJF
Choose: 1
Enter overs (23-33): 33

Using RR:
Bowler Sequence:
Player IDs: 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8

Bowlers Stats:
Pid    Runs    W    Balls
7      21      2    49
3      23      2    49
2      25      2    49
8      28      2    49

Using SRTF:
Batter Sequence:
Player IDs: 6 5 5 4 1 1

Batters Stats:
Pid    Runs    Balls    Out?
6      19      34      1
5      10      13      1
4      24      44      1
1      21      34      1

Do you want to update the new data? (Y/N) : n
```

```
Choose the scheduling type for batter:
1. SRTF
2. SJF
Choose: 2
```

```
Choose the scheduling type for bowler:
1. RR
2. SJF
Choose: 2
Enter overs (23-33): 33
```

```
Using SJF:
Bowler Sequence:
Player IDs: 7 3 2 8
```

```
Bowlers Stats:
Pid    Runs    W    Balls
7      40      2    49
3      41      2    49
2      43      2    49
8      46      2    49
```

```
Using SJF:
Batter Sequence:
Player IDs: 4 1 6 5
```

```
Batters Stats:
Pid    Runs    Balls    Out?
4      24      44      1
1      21      39      1
6      19      34      1
5      10      16      1
```

```
Do you want to update the new data? (Y/N) : n
```

Bowlers

Runs_Conceded	Balls_Bowled	Wickets_Taken
26	33	2
29	37	2
21	30	2
23	34	2
13	30	2
10	22	2
13	30	2
7	15	2
7	16	2
21	49	2

Batters

Runs_Scored	Balls_Faced	Is_Dismissed
12	22	1
12	22	1
12	22	1
19	34	1
11	20	1
11	20	1
11	20	1
7	12	1
10	17	1
19	34	1

Predicted Run Scores and Wickets

Bowlers

Using RR

Bowler Sequence:

Player IDs: 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8 7 3 2 8

Bowlers Stats:

Player ID	Runs Conceded	Wickets Taken	Balls Bowled
7	21	2	49
3	23	2	49
2	25	2	49
8	28	2	49

Using SJF

Bowler Sequence:

Player IDs: 7 3 2 8

Bowlers Stats:

Player ID	Runs Conceded	Wickets Taken	Balls Bowled
7	40	2	49

3	41	2	49
2	43	2	49
8	46	2	49

Batters

Using SRTF

Batter Sequence:

Player IDs: 6 5 5 4 1 1

Batters Stats

Player ID	Runs Scored	Balls Faced	Out?	Strike Rate
6	19	34	1	62.5
5	10	13	1	55.88
4	24	44	1	53.84
1	21	34	1	54.54

Using SJF

Batter Sequence:

Player IDs: 4 1 6 5

Batters Stats

Player ID	Runs Scored	Balls Faced	Out?	Strike Rate
4	24	44	1	65.21
1	21	39	1	65
6	19	34	1	68.75
5	10	16	1	75

Comparison of Each Scheduling Strategy

Bowlers

Using RR

The players keep changing their turns after each over, so they keep up their efficiency and do not get exhausted. That's why their runs conceded per ball remain constant.

Using SJF

The players bowl constantly until all their specified number of balls are finished. This reduces their efficiency as they get exhausted bowling constantly.

Batters

Using SRTF

The batters with the shortest number of balls left are brought on strike first because they are in their form. This leads to an increase in their performance, and they score their target runs in a smaller number of balls, increasing their strike rate.

Using SJF

The batters who have the highest strike rate are brought on strike, and they play until they are dismissed.

Recommendations

Bowlers

For bowlers, it is recommended to use Round Robin, as this allows their runs conceded per ball constant, and maintains their efficiency.

Batters

For batters, the Shortest Remaining Time First is recommended, as the players in good form will perform first and score more in a smaller number of balls, increasing their strike rates and improving their career statistics.